Chapter 8. Bootstrap Confidence Intervals

Normal, standard, bootstrap-t, percentile and BCa intervals

Contents

Setting

From bootstrap replicates $\hat\theta^{*1},\dots,\hat\theta^{*B}$ (distribution G^*, quantiles Q_{G^*}) we build a $1-\alpha$ interval for θ. The methods differ in what they assume is (approximately) a pivot.

8.1 Normal and basic intervals

Normal-approximation bootstrap interval. If $(\hat\theta-\text{bias}-\theta)/\,\text{se}\,\dot\sim\,N(0,1)$,

$$\theta \in \hat\theta - \widehat{\text{bias}}_{\text{Boot}} \pm z_{\alpha/2}\widehat{\text{se}}_{\text{Boot}}.$$

It uses only two summaries of G^* and is only as good as the normal approximation.

Basic interval. Assume $\hat\theta-\theta\sim H$ with H free of θ (a pivot) and approximate H by the law H^* of $\hat\theta^*-\hat\theta$, whose quantiles are $Q_{G^*}(p)-\hat\theta$. Since $P(Q_H(\frac{\alpha}{2})\le\hat\theta-\theta\le Q_H(1-\frac{\alpha}{2}))=1-\alpha$,

$$\theta\in\big(2\hat\theta-Q_{G^*}(1-\tfrac{\alpha}{2}),\ 2\hat\theta-Q_{G^*}(\tfrac{\alpha}{2})\big). \tag{8.1}$$

8.2 The bootstrap-t interval

Assume instead that the studentized quantity $t=(\hat\theta-\theta)/\widehat{\text{se}}$ is (approximately) pivotal, and approximate its law by that of

$$t^*=\frac{\hat\theta^*-\hat\theta}{\widehat{\text{se}}^*},$$

where $\widehat{\text{se}}^*$ is the standard error estimate recomputed on each bootstrap sample. With Q_{t^*} the quantiles of the t^{*b},

$$\theta\in\big(\hat\theta-\widehat{\text{se}}\,Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\theta-\widehat{\text{se}}\,Q_{t^*}(\tfrac{\alpha}{2})\big). \tag{8.2}$$

If $\widehat{\text{se}}$ has a formula (e.g. $s/\sqrt{n}$ for a mean, or regression standard errors), this is cheap. Otherwise each $\widehat{\text{se}}^*$ needs a second-level bootstrap ($B\times C$ resamples, with $C\approx 25$ usually enough). The method works best on a scale where t is close to pivotal. E.g. for a variance, apply it to $\log\hat\theta$ and exponentiate the endpoints; an increasing transformation preserves coverage.

8.3 Percentile intervals

Motivation. The bootstrap distribution of an estimator approximates its sampling distribution, so its quantiles give a confidence interval without any formula for the standard error. This section studies when the simple percentile interval works.

Percentile interval. $\theta\in(Q_{G^*}(\frac{\alpha}{2}),Q_{G^*}(1-\frac{\alpha}{2}))$.

Proposition 8.1

Suppose there is an increasing φ, not necessarily known, with $\varphi(\hat\theta)-\varphi(\theta)\sim N(0,1)$, and the same holds in the bootstrap world: $\varphi(\hat\theta^*)-\varphi(\hat\theta)\sim N(0,1)$ given the data. Then the percentile interval is exact.

Proof. The exact interval is $\varphi(\theta) \in \varphi(\hat\theta) \pm z_{\alpha/2}$, i.e. $\theta \in (\varphi^{-1}(\varphi(\hat\theta) - z_{\alpha/2}), \varphi^{-1}(\varphi(\hat\theta) + z_{\alpha/2}))$. In the bootstrap world $\varphi(\hat\theta^*) \sim N(\varphi(\hat\theta), 1)$, so $\varphi(\hat\theta) \pm z_{\alpha/2}$ are its $\frac{\alpha}{2}$ and $1-\frac{\alpha}{2}$ quantiles. Since φ is increasing, applying φ^{-1} maps them to the quantiles $Q_{G^*}(\frac{\alpha}{2})$, $Q_{G^*}(1-\frac{\alpha}{2})$ of $\hat\theta^*$. The unknown φ never has to be computed. ■

The percentile interval is transformation-respecting and always lies in the parameter space. Its weak point is the assumption that $\varphi(\hat\theta)$ is median-unbiased with constant variance.

8.3.1 Bias-corrected and accelerated (BCa) intervals

Generalize the assumption to

$$\varphi(\hat\theta) - \varphi(\theta) \sim N(-b\,\sigma_\varphi,\ \sigma_\varphi^2), \qquad \sigma_\varphi = 1 + a\,\varphi(\theta), \tag{8.3}$$

with a bias constant b and an acceleration a (the rate at which the standard error changes with the parameter).

Proposition 8.2

Under (8.3) and its bootstrap analogue, the interval $(Q_{G^*}(p_l), Q_{G^*}(p_u))$ is exact, where with $z_l = \Phi^{-1}(\frac{\alpha}{2})$, $z_u = \Phi^{-1}(1-\frac{\alpha}{2})$,

$$p_l = \Phi\Big(b + \frac{b + z_l}{1 - a(b + z_l)}\Big), \qquad p_u = \Phi\Big(b + \frac{b + z_u}{1 - a(b + z_u)}\Big).$$

Proof [Supplement]. Write $\hat\varphi = \varphi(\hat\theta)$, $\phi = \varphi(\theta)$, and assume $1 + a\phi > 0$. Then $Z = \frac{\hat\varphi - \phi}{1+a\phi} + b \sim N(0,1)$. Solving, $\phi = g(Z) = \frac{\hat\varphi - (Z-b)}{1+a(Z-b)}$ with $g'(Z) = -\frac{1+a\hat\varphi}{(1+a(Z-b))^2} < 0$. So $\phi \in [g(-z_l), g(-z_u)] = [L, U]$ iff $Z \in [z_l, z_u]$, which has probability $1-\alpha$; here $L = \frac{\hat\varphi + b + z_l}{1-a(b+z_l)}$ and similarly U. In the bootstrap world, $\varphi(\hat\theta^*) \sim N(\mu_*, \sigma_*^2)$ with $\mu_* = \hat\varphi - b(1+a\hat\varphi)$, $\sigma_* = 1 + a\hat\varphi$. With $w = b + z_l$, $L - \hat\varphi = \frac{w(1+a\hat\varphi)}{1-aw}$, so $P^*(\varphi(\hat\theta^*) \le L) = \Phi(\frac{L-\mu_*}{\sigma_*}) = \Phi(\frac{w}{1-aw} + b) = p_l$. Thus $\varphi^{-1}(L) = Q_{G^*}(p_l)$, and similarly for U. ■

Estimating b and a. Under the model, $P^*(\hat\theta^* \le \hat\theta) = \Phi(b)$, so

$$\hat b = \Phi^{-1}\Big(\frac{\#\{\hat\theta^{*b} \le \hat\theta\}}{B}\Big).$$

For a, use the jackknife: with $\hat\theta_{(i)}$ computed without observation i and $\hat\theta_{(\cdot)}$ their mean,

$$\hat a = \frac{\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^3}{6\big[\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^2\big]^{3/2}},$$

a skewness measure of the influence of each observation. The jackknife (leave-one-out recomputation; Quenouille 1949, Tukey 1958) was originally proposed for bias and standard-error estimation and is a precursor of the bootstrap. With $a = 0$ the method reduces to the **bias-corrected (BC)** interval, with $p = \Phi(2\hat b + z)$.

8.4 Accuracy and recommendations

For a smooth statistic, the coverage of an interval C satisfies $P(\theta \in C) = 1 - \alpha + O(n^{-k/2})$:

- first-order accurate ($k = 1$): normal theory, normal bootstrap, basic, percentile, BC;
- second-order accurate ($k = 2$): bootstrap-t, BCa.

Recommendation: bootstrap-t when a reliable standard error formula exists (it then needs no second-level resampling); otherwise BCa. Use $B \geq 2000$ for intervals, since tail quantiles are needed. The percentile interval is the simplest default and is often adequate for large samples. For multi-sample or regression resampling, the jackknife estimate of a is less reliable.

8.5 Selected exercises

Motivation. The exercises show that a normalizing transformation behind the percentile-type intervals always exists, compare how the basic and percentile intervals behave under reparametrization, and compute the jackknife for two simple statistics.

Exercise 8.2 (A normalizing transformation always exists)

Let $\hat{\theta}$ have a continuous, strictly increasing distribution function G (for the true θ), and put $\varphi(t) = \Phi^{-1}(G(t))$.
(a) Show that φ is increasing and $\varphi(\hat{\theta}) \sim N(0, 1)$.
(b) If θ is the median of G, show that $\varphi(\hat{\theta}) - \varphi(\theta) \sim N(0, 1)$.
(c) In general, with $b = \varphi(\theta)$, show that $\varphi(\hat{\theta}) - \varphi(\theta) \sim N(-b, 1)$.

Solution.

(a) φ is a composition of the increasing maps G and Φ^{-1}. By Exercise 4.6(a), $U = G(\hat{\theta}) \sim$ Uniform$(0, 1)$, and then $P(\Phi^{-1}(U) \leq z) = P(U \leq \Phi(z)) = \Phi(z)$.

(b) $G(\theta) = \frac{1}{2}$ gives $\varphi(\theta) = \Phi^{-1}(\frac{1}{2}) = 0$; apply (a).

(c) $\varphi(\hat{\theta}) - \varphi(\theta) = Z - b$ with $Z = \varphi(\hat{\theta}) \sim N(0, 1)$.

So the assumption of Proposition 8.1 (case (b)) and that of the BC interval, (8.3) with $a = 0$ and $\sigma_\varphi = 1$ (case (c)), hold for some increasing φ whenever the sampling distribution is continuous. The real restriction lies elsewhere: φ is built from G, which depends on θ and F, whereas the propositions require one φ that works both for the true θ and in the bootstrap world. ■

Exercise 8.7 (Basic versus percentile under reparametrization)

(a) Let g be a continuous, strictly increasing bijection between intervals containing the possible values of $\hat{\theta}$ and $\hat{\theta}^*$. Show that the percentile interval for $g(\theta)$, computed from the replicates $g(\hat{\theta}^{*b})$, equals g applied to the percentile interval for θ.
(b) Use the exact bootstrap distribution of Example 7.1 (data $\{1, 4, 9\}$, sample median) to compute the 60% basic and percentile intervals for the population median θ, and the basic interval for $\sqrt{\theta}$. Show that the basic interval does not have the property in (a).

Solution [Supplement].

(a) Let T have distribution function F_T. For every x, $P(g(T) \leq g(x)) = P(T \leq x)$, so $F_{g(T)}(g(x)) = F_T(x)$. Since g is an increasing bijection,

$$Q_{g(T)}(p) = \min\{g(x) : F_T(x) \geq p\} = g(\min\{x : F_T(x) \geq p\}) = g(Q_T(p)).$$

Apply this with $p = \frac{\alpha}{2}$ and $1 - \frac{\alpha}{2}$ to the bootstrap distribution G^*.

(b)

Step 1. G^* puts mass $7/27, 13/27, 7/27$ on $1, 4, 9$, and $\hat\theta = 4$. With $\alpha = 0.4$: $G^*(1) = 0.26 \geq 0.2$, so $Q_{G^*}(0.2) = 1$; $G^*(4) = 0.74 < 0.8$, so $Q_{G^*}(0.8) = 9$.

Step 2. Percentile interval for θ: $(1, 9)$. Basic interval (8.1): $(2 \cdot 4 - 9,\ 2 \cdot 4 - 1) = (-1, 7)$.

Step 3. For $\psi = \sqrt{\theta}$: $\hat\psi = 2$, and the replicates are $1, 2, 3$ with the same probabilities. The percentile interval is $(1, 3)$, i.e. $(1, 9)$ for θ, in agreement with (a). The basic interval is $(2{\cdot}2{-}3,\ 2{\cdot}2{-}1) = (1, 3)$, which maps back to $(1, 9) \neq (-1, 7)$.

The basic interval thus depends on the scale on which it is computed. On the original scale it reflects the skewed bootstrap distribution the wrong way and even reaches negative values, which are impossible if the population is known to be positive. ■

Exercise 8.8 (The jackknife for two statistics)

Let $\hat\theta_{(i)}$ be the statistic computed without observation i and $\hat\theta_{(\cdot)} = \frac{1}{n}\sum_i \hat\theta_{(i)}$. The jackknife bias-corrected estimator and standard error are

$$\hat\theta_{\text{jack}} = n\hat\theta - (n-1)\hat\theta_{(\cdot)}, \qquad \widehat{\text{se}}_{\text{jack}} = \Big(\frac{n-1}{n}\sum_i (\hat\theta_{(i)} - \hat\theta_{(\cdot)})^2\Big)^{1/2}.$$

(a) For $\hat\theta = \hat\sigma^2 = \frac{1}{n}\sum_i (X_i - \bar X)^2$, show that $\hat\theta_{\text{jack}} = s^2$.
(b) For $\hat\theta = \bar X$, show that $\widehat{\text{se}}_{\text{jack}} = s/\sqrt{n}$.

Solution [Supplement]. Let $SS = \sum_j (X_j - \bar X)^2$ and $\bar X_{(i)} = (n\bar X - X_i)/(n-1)$, so that $\bar X_{(i)} - \bar X = (\bar X - X_i)/(n-1)$.

(a)

Step 1. (Deleting one point.) By the decomposition about $\bar X_{(i)}$,

$$SS_{(i)} := \sum_{j\neq i}(X_j - \bar X_{(i)})^2 = \sum_{j\neq i}(X_j - \bar X)^2 - (n-1)(\bar X_{(i)} - \bar X)^2$$
$$= SS - (X_i - \bar X)^2 - \frac{(X_i - \bar X)^2}{n-1} = SS - \frac{n}{n-1}(X_i - \bar X)^2.$$

Step 2. So $\hat\theta_{(i)} = SS_{(i)}/(n-1)$, and summing over i,

$$\hat\theta_{(\cdot)} = \frac{1}{n(n-1)}\Big(nSS - \frac{n}{n-1}SS\Big) = \frac{(n-2)SS}{(n-1)^2}.$$

Step 3. Hence $\hat\theta_{\text{jack}} = SS - \frac{(n-2)SS}{n-1} = \frac{SS}{n-1} = s^2$. The jackknife removes the bias of $\hat\sigma^2$ exactly.

(b) $\hat\theta_{(i)} = \bar X_{(i)}$, so $\hat\theta_{(\cdot)} = \bar X$ and $\hat\theta_{(i)} - \hat\theta_{(\cdot)} = (\bar X - X_i)/(n-1)$. Then

$$\widehat{\text{se}}^2_{\text{jack}} = \frac{n-1}{n}\cdot\frac{SS}{(n-1)^2} = \frac{SS}{n(n-1)} = \frac{s^2}{n}.$$

The jackknife reproduces the usual standard error exactly; the ideal bootstrap gives the slightly smaller $\hat\sigma/\sqrt{n}$ (Exercise 7.5). ■